

09 QUESTIONS: the tagged MTE question inventory

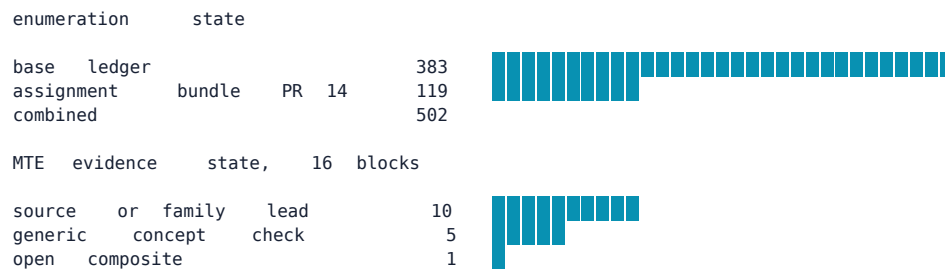
THE QUESTION INVENTORY AS A GRID

the two MTE papers (16 blocks)					
M24	A1 A2 A3 B1 B2 B3 B4 C1(OPEN)				
M25	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8				
10 source/family leads	5 concept checks	1 open composite			
assignments (A1 + A2 = 52 blocks)					
A1	10 MCQ	6 short	4 long	4 application	
A2	12 A	8 B	4 C	4 D	
the teaching decks (60 blocks)					
notes	30 blocks (D1-1 .. D1-30)				
ppt3	6	ppt4	7	clt	5 ppt5 5
lms-th	7 (2 unique)				
re-teach decks (30 blocks, pass 2)					
L1-7	13	L8-9	4	L12-13 6	L14-15 7

THE SLOT COVERAGE MAP (what is asked vs what is one mutation away)

SLOT:	1	2	3	4	5 6 7 8 9
pt tl iv mo pa in ct cd co					
BINOMIAL		X	X	S	X X S X S S
POISSON		X	X	.	X X . X . X
UNIFORM		X	X	X	X S S . . S
NORMAL		X	X	X	X X X X . X
EXPONENTIAL	X	X	X	X	X . . X X
X = asked in our corpus					
S = a sibling, one mutation away, NOT yet asked	<- risk				
. = no evidence					
the S cells are where the next paper can surprise you.					
the four highest-risk siblings: normal one-unknown,					
binomial interval, CLT-for-a-sum, discrete conditional.					

PROVENANCE TRUTH, VISUALISED



The 502 count measures enumerated question instances, not unique questions and not sourced questions. No content-family ids have been assigned yet, so corpus-wide deduplication and source attribution remain open work.

Every MTE-scope question block we hold, tagged by distribution, slot (the nine in 07-PATTERNS), and where it lives. This is the drill list. "In scope" means lectures 1-21.

The nine slots: 1 point, 2 tail, 3 interval, 4 moments, 5 parameter, 6 inverse, 7 count, 8 conditional, 9 compose.

A. The two MTE papers (16 blocks, evidence states kept separate)

id	source page	question	dist	slot	status
M24-A1	mte 2024 p001	Poisson transform concept	Poi	4	generic concept
M24-A2	mte 2024 p001	triangular density, mean, variance	contin	4+5	traced G&K 8-1-5
M24-A3	mte 2024 p001	Chebyshev forms, pick the false ones	Cheb	-	concept
M24-B1	mte 2024 p001	$\text{Var}(X-2Y)$, λ^2 , μ^3 , answer 14	Poisson	4	VERBATIM G&K
M24-B2	mte 2024 p001	$f(x)=Ae^{-x/5}$ telephone	Exp	4	reskin G&K ch5 ex10
M24-B3	mte 2024 p001	Chebyshev interval $(-1,1)$ vs $(-1,3)$	Cheb	2	reskin G&K ex15b
M24-B4	mte 2024 p001	uniform wait, answer $1/3$	Unif	3	G&K subway
M24-C1	mte 2024 p002	rain $N(2.6,34.5)$ + $5/3$ binomial week	Normal+Bin 9	OPEN (composite)	
M25-Q1	mte 2025 p001	axiom of total mass	prob	-	concept
M25-Q2	mte 2025 p001	substitution drill	prob	-	concept
M25-Q3	mte 2025 p001	sufficient statistics MCQ	Est	-	concept cross-check
M25-Q4	mte 2025 p001	density $kx^3(4-x)^2$, find k , mean, var	contin	4+5	ABES sample (asks SD)
M25-Q5	mte 2025 p001	Chebyshev find c , μ 10, var 4, bound .04	Cheb	6	deck L10-11 Q2(iv)
M25-Q6	mte 2025 p001	exponential reskin, mean 5 $\rightarrow$ 3	Exp	4	G&K ch5 ex10 reskin
M25-Q7	mte 2025 p001	machine life sample mean, $n=9$	Normal	3	Walpole P&S 8.25
M25-Q8	mte 2025 p002	(i) Poisson notes (ii) estimation chapter	Poi+Est	9	mixed

B. Assignment 1 (24 blocks) and Assignment 2 (28 blocks)

id	question	dist	slot
A1 MCQ1	3 children, coin, pmf and cdf	discrete	1+3
A1 MCQ5	Chebyshev applicability	Cheb	2
A1 MCQ10	Chebyshev applicability	Cheb	2
A1 short1	pmf $1/8 \ 3/8 \ 3/8 \ 1/8$ + cdf	discrete	3
A1 short2	mortality pdf $P(X>12) = 0.0916$	contin	2
A1 short3	$f(x)=6x(1-x)$, valid pdf, $b=1/2$	contin	5+1
A1 short4	geometric expectation, $E=2$	geom(gap)	4
A1 short5	dice Chebyshev $35/54$ vs $1/3$	Cheb	9
A1 short6	600 throws, $19/24$	Bin+Cheb	9
A1 long1	mortality integral + conditional 0.4863	contin	3+8
A1 long2	pmf with unknown k , $k=1/10$, $P(X<6)=0.81$	discrete	5+3
A1 long3	lot of 25, 5 defective, mean 0.8 both ways	Hyper(no slide!) 7	
A1 long4	four bad oranges $12/19$, $32/95$, $3/95$	discrete	3
A1 app1	sensor batch, $E=1.2$, $\text{Var}=0.86$	discrete	4
A1 app2	CDF $0.15, 0.50, 0.80, 1.00$	discrete	4
A1 app3	battery $E=3.45$, $\text{Var}=0.9475$, $\text{sd}=0.97$	discrete	4
A1 app4	marks Chebyshev, 75 percent	Cheb	9
A2 A2	binomial moments	Bin	4
A2 A3	Poisson	Poi	1
A2 A4	uniform E , Var	Unif	4
A2 A7	Poisson	Poi	see source
A2 A9	binomial max successes	Bin	4
A2 A10	uniform parameters	Unif	5
A2 A11	normal $68/95/99.7$ landmarks	Normal(no slide!) 4	
A2 A12	Poisson additivity	Poi(no slide!) 9	
A2 B1	Poisson rejected $P(X\geq 4)=0.2424$	Poi	2
A2 B2	uniform	Unif	see source
A2 B3	normal interval 45-62	Normal	3
A2 B4	5000 batteries $N \times P$	Normal	7
A2 B5	exponential conditional 0.6225	Exp(no slide! 8)	8
A2 B8	uniform $ X $ intervals	Unif	3
A2 C1	exponential memoryless	Exp(no slide!) 8	
A2 C2	two-unknown normal $37.2/28.2/30.4$	Normal	5+6
A2 C3	Poisson $P(X>2)=0.8753$	Poi	2
A2 C4	uniform rounding	Unif	1
A2 D1	10000 bulbs $N \times P$	Normal	7
A2 D2	exponential unit conversion	Exp	5
A2 D3	binomial moments	Bin	4

id	question	dist	slot
A2 MCQ5	exponential memoryless	Exp	8
A2 MCQ11	normal landmarks	Normal	4

C. The teaching decks' own questions (60 blocks, the practice pool)

notes deck (30 blocks, p041 to p147)	see 10-SLIDE-MAP.md for the page-by-page list
ppt3 (6 blocks)	5-coin, pens x3, irregular die x2
ppt4 (7 blocks)	uniform, normal chain, inverse, exponential
clt (5 blocks)	lightbulbs, ATM, impurity, LED, machine
ppt5 (5 blocks)	response time, packets, battery CI, comparison set 1, set 2
lms-theory (7 total, 2 unique)	sufficiency examples

D. The re-teach deck questions (30 blocks, use as pass 2)

L1-7	13 blocks: L17-01..L17-13 (faculty computers, basketball, camera, rare disease,
gas pumps, cylinder pmf [OPEN], county forms, batteries, boards, dice max, cdf proof,	
E on -3/6/9)	
L8-9	4 blocks: L89-01..L89-04 (bus pdf, CDF-to-density [OPEN], kidney E(Y), Pareto)
L12-13 6 blocks: L1213-01..06 (5-coin [OPEN], pens, irregular die, Poisson x3 [one OPEN])	
L14-15 7 blocks: L1415-01..07 (uniform, normal with mean 8 and sd 5, inverse cutoff, exponential	
McClave)	

E. The open items (no source found yet)

M24-C1	rain set + 5/3 binomial composite	in scope, no source
L17-06	pmf table with k and CDF questions	four-deck, open
L17-13	pmf on -3/6/9, transformed expectation	four-deck, open
L89-02	CDF to density	four-deck, open
L1213-01 exactly 3 heads in 5 tosses	four-deck, open	
L1213-03 irregular die even outcomes	four-deck, open	
L1213-04 Poisson relation $P_1 = 0.2 P_2$	four-deck, open	
L1213-05 Poisson calls exactly two each minute	four-deck, open	
L1213-06 life-insurance Poisson approximation	four-deck, open	
(the pens question circulates in several versions, but its originating source is OPEN)		

F. Provenance state (honest, as of 16 Sep)

Enumeration: 383 rows on this branch, plus 119 assignment-bundle rows on PR 14.	
Combined after merge: 502 gross instances.	
MTE evidence: 10 source or family leads, 5 generic concept checks, 1 open composite.	
A concept check confirms mathematical type, not copying or source provenance.	
Corpus-wide source matching is incomplete. No row carries a content family id yet, so	
neither the unique-question count nor a corpus-wide sourced percentage is known.	